Statistics

Chapter 12 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 10

Chapter 12 **Exercise 1** Page 459

PROBLEM STATEMENT

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Solution A

Step 1

The null hypothesis and the alternative hypothesis both state a claim about a population parameter.

However, the **null hypothesis** always needs to include an equality.

Result

Null hypothesis

Chapter 12 Exercise 2

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PROBLEM STATEMENT

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Solution A

Step 1

The **significance level** is the probability of rejecting the null hypothesis $H_.$, when the null hypothesis H_0 is true.

Result

Significance level

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PROBLEM STATEMENT

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Solution A

Step 1

It is only appropriate to conduct a hypothesis test about a population proportion, when the number of successes in the sample is at least 5.

Since the sample proportion π is the number of successes divided by the sample size n, the value $n\pi$ represents the number of successes and thus the value $n\pi$ needs to be at least $\mathbf{5}$.

Result

5

Chapter 12 Exercise 4

PROBLEM STATEMENT

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Solution A

Step 1

When performing a hypothesis test about a single population mean, we need to use the z-distribution when the **population standard deviation** σ is known and we need to use the t-distribution when the population standard deviation σ is unknown.

Result

Population standard deviation

Chapter 12 Exercise 5

PROBLEM STATEMENT

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Solution A

Step 1

When conducting a two-sample hypothesis test about means with the population standard deviations unknown, we require that the population distributions are **approximately normally distributed** in order to use the t-distribution in the hypothesis test.

Result

Approximately normal

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PROBLEM STATEMENT

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Solution A

Step 1

The value of the **test statistic** is determined by using sample information and is used to determine whether to reject the null hypothesis H_0 or not.

Result

Test statistic

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PROBLEM STATEMENT

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Solution A

Step 1

In a two-tailed hypothesis test, we need to take into account both tails of the distributions and thus the rejection region will be **evenly split between the two tails**.

Result

Evenly split between the two tails

PROBLEM STATEMENT

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Solution A

Step 1

The F-distribution is a continuous probability distribution, which is positively skewed.

Moreover, the F-distribution represents a family of distribution, because different values for the degrees of freedom for the numerator and/or degrees of freedom for the denominator result in a different F-distribution.

However, the F-distribution does not range from $-\infty$ to $+\infty$, because the distribution only ranges from 0 to $+\infty$.

Result

Range from $-\infty$ to $+\infty$

PROBLEM STATEMENT

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Solution A

Step 1

Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

Two events are **disjoint** or **mutually exclusive**, if the events cannot occur at the same time.

When performing a one-way ANOVA test, we require that the treatments are *independent* such that one treatment won't influence other treatments.

Result

Independent

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 4 \\ n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 6 \\ n_3 &= \text{Sample size} = 6 \\ n_4 &= \text{Sample size} = 6 \\ n &= n_1 + n_2 + n_3 + n_4 = 6 + 6 + 6 + 6 = 24 \end{aligned}$$

The degrees of freedom for the numerator is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

The degrees of freedom for the denominator is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 4 = 20$$

Result

3 and 20